HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained the **Keet Desktop** application using **React** and **TypeScript**, focusing on UI/UX implementation and cross-platform compatibility (macOS, Windows, Linux).
- Implemented robust state management patterns using **Redux Toolkit** to handle complex application states and ensure predictable data flow across the desktop application.
- Wrote comprehensive unit and integration tests for frontend components and application logic using **Jest** and **React Testing Library**, achieving over 85% code coverage.
- Collaborated with a globally distributed team on the **Pear runtime**, contributing to the development of P2P communication features and direct application deployment capabilities.
- Optimized application performance by refactoring rendering logic and improving memory management in the **Keet Desktop** application, reducing CPU usage by approximately **18%** during peak load.
- Integrated real-time communication features, including basic **WebRTC** signaling for peer discovery and connection establishment within the P2P framework.
- Containerized the desktop application build process using **Docker** for consistent development and deployment environments across different operating systems.

SKILLS

Frontend – React, Preact, JavaScript, TypeScript, Redux, Redux Toolkit, Zustand, WebRTC, HTML, CSS

Desktop App Dev – Electron, Tauri, macOS, Windows, Linux

P2P & Distributed Systems – BitTorrent, Pear runtime, Node.js

Testing & Debugging – Jest, React Testing Library, Debugging

Cloud & DevOps – Docker, CI/CD

PROJECTS

Keet Desktop P2P Communications App

- Developed the core user interface and client-side logic for a cross-platform desktop messaging application using **React** and **TypeScript**, designed for peer-to-peer communication.
- Integrated state management solutions with **Redux Toolkit** to manage user sessions, contact lists, and message history efficiently, ensuring data consistency.
- Implemented testing strategies with **Jest** and **React Testing Library** to validate component functionality and application workflows, leading to fewer production issues.

Pear Runtime P2P Integration Module

- Contributed to the development of a Node.js-based module for the **Pear runtime**, focusing on establishing and managing direct peer-to-peer connections.
- Engineered data replication mechanisms to ensure data consistency across distributed nodes, drawing parallels to **BitTorrent** principles.
- Designed and implemented RESTful APIs for inter-service communication within the decentralized architecture, facilitating seamless application deployment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)